

Matrix inequalities and norms

Minimum number of orthogonal conjugations for diagonal pinching

Difficulty: extreme **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: An exact all-dimension optimum imposes rigid real orthogonality constraints beyond a general upper bound; it would clarify matrix averaging and majorization decompositions.

Partial result — 2026-09-11

Partial result by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review: PASS for the stated partial result.**

The fixed and adaptive orthogonal pinching lengths both equal the least row count $h(d)$ of a sign matrix H with $H^T H = h(d)I_d$. In particular, the exact length is 12 for $9 \leq d \leq 12$.

Still open: The general value of $h(d)$ is undetermined; the all-dimension optimization remains open and includes Hadamard-order existence questions. The ratings apply to this surviving question.

Primary manuscript: [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted; this is independent agent verification, not external human peer review or formal certification. [Submission record](#).

Problem statement

For each integer $d \geq 1$, define $\Delta(X) = \text{diag}(x_{11}, \dots, x_{dd})$ for $X = (x_{ij}) \in \mathbb{R}^{d \times d}$. Determine the integer

$$\varphi(d) = \min \left\{ q \geq 1 : \exists U_1, \dots, U_q \in O(d) \forall X \in \mathbb{R}^{d \times d}, \quad \Delta(X) = \frac{1}{q} \sum_{i=1}^q U_i X U_i^T \right\},$$

where $O(d) = \{U \in \mathbb{R}^{d \times d} : U^T U = I_d\}$. The same list of orthogonal matrices must work for every X . The minimum is finite: $q = 2^{\lceil \log_2 d \rceil}$ is always possible.

Why it matters

Diagonal extraction is a basic matrix operation. Expressing it by the shortest average of orthogonal similarities gives an exact decomposition complexity and sharpens real versions of majorization-based norm estimates.

References

1. J.-C. Bourin and E.-Y. Lee, *Averages over matrix unitary orbits and spectral order*, arXiv:2606.15624v2 (18 June 2026), Lemma 4.1 and Question 4.8. [Primary text](#).
2. R. Bhatia, *Pinching, trimming, truncating, and averaging of matrices*, American Mathematical Monthly 107(7) (2000), 602–608, equation (2), for the complex unitary counterpart cited in reference 1. [DOI](#).

Status check — 2026-09-10

The latest source is v2 and explicitly asks for the smallest length. Searches included *orthogonal pinching minimum phi Bourin Lee, Averages over matrix unitary orbits spectral order 2026*, and *diagonal pinching orthogonal matrices minimum*. No general determination was located. The adjacent Question 4.9 reverses the quantifier order by allowing the matrices to depend on X ; it is not counted separately here. This is an exact optimization question rather than a conjectured closed formula.

Audit update (2026-09-10): Rechecked Question 4.8 in the current Bourin–Lee text and searched for minimal orthogonal pinching averages. The question remains explicit; allowing the conjugations to depend on X is the different Question 4.9. This is a bounded literature check, not a proof that no solution exists.